(KTXFI1EBNF) Physics I. Lecture

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Current Updates and Announcements

- The exam dates and locations (rooms) have been finalized.
- Opportunities to retake signatures will be available on May 26, 2026.
- New mechanics exercises will be posted $\rightarrow$ Homework No. 4 and 5.

Where are we? I

1. **Mechanics I.:** Newton's laws, equation of motion (review). Momentum. General concept of mechanical work, work theorem. Forms of mechanical energy.
2. **Mechanics II.:** Weight force, gravitational force, gravitational field. Work of the gravitational force. Conservative force field. Dissipative force field. Mechanical power.
3. **Mechanics of Rigid Bodies I.** Concept of a rigid body. Forces acting on a rigid body. Motion of a rigid body. Concept of the centre of mass. Location of the centre of mass. Torque. Angular momentum. Energy of a rotating rigid body, angular momentum of a rotating rigid body, Steiner's theorem.
4. **Mechanics of Rigid Bodies II.** Principal theorems of rigid body motion; conservation laws. Collisions.
5. **Oscillations I.** Dynamic condition of harmonic oscillatory motion. Differential equation of harmonic undamped oscillatory motion.

Where are we? II

6. **Oscillations II.** Superposition of oscillations. Decomposition of oscillations into components. Damped oscillations. Forced oscillations, resonance. Coupled oscillations.
7. **Wave phenomena:** Reflection of waves, refraction of waves, diffraction of waves. Huygens–Fresnel principle. Interference of waves. Polarization of waves.
8. **Elements of acoustics:** Basics of acoustics. Fundamental concepts. Doppler effect. Sound intensity, loudness according to Weber–Fechner's law.
9. **Thermodynamics I:** Temperature scales, thermal expansion, thermometers. State variables, ideal gas, state changes, ideal gas equation of state, Boyle's law, Gay-Lussac's laws, universal (combined) gas law, heat, specific heat, molar heat, heat capacity.
10. **Thermodynamics II:** Expansion work, internal energy, the first law of thermodynamics, forms of the first law in different state changes.

Where are we? III

11. **Thermodynamics III:** Cyclic processes, Carnot cycle, Carnot's theorem, reduced heat, entropy, the second law of thermodynamics.
12. **Thermodynamics IV:** Kinetic theory of gases, molecular thermodynamics. Distributions, distribution functions. Elementary statistics.
13. **Electrostatics I.:** *Electric charge: Fundamental electrical phenomena, electric charge and matter, Coulomb's law (force between two point charges, unit of electric charge, permittivity of vacuum).*
14. **Electrostatics II.:** *The electric field: The electrostatic field (electric field strength, electric field lines). Determination of electric field strength. Gauss's law (flux of the electric field). Electric field of a point charge.*

Where are we? IV

15. **Electrostatics III.:** Electrostatic potential: Work of the electric field (work done in an electric field). Electric potential (electrostatic potential). Electric potential difference, voltage. Electric potential of charge systems at small and large distances. Electrostatic energy.
16. **Elements of physical or wave optics.** Branches of optics. Light interference, light diffraction, diffraction through a slit, optical grating, light polarization. Light as an electromagnetic wave: Poynting vector. Optical instruments.

Fundamental Electrical Phenomena I

- **Observation:** Certain materials, when rubbed together, exert forces on each other (electrostatics).
- **Two types of charges:** Benjamin Franklin named them *positive* and *negative*.
- **Basic Rule:**
  - Like charges **repel** each other.
  - Opposite charges **attract** each other.
- **Conservation of Charge:** The total electric charge in an isolated system remains constant.

Fundamental Electrical Phenomena II

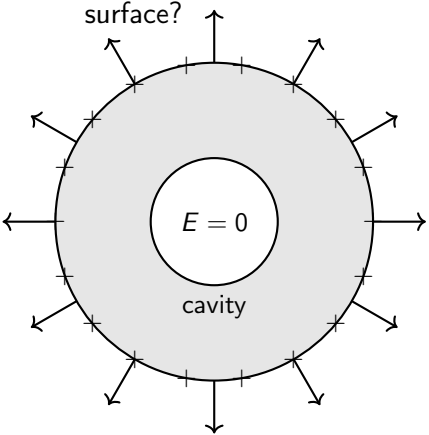
- Electric Charge and Matter
  - **Microscopic View:** Matter is composed of atoms.
    - **Protons:** Positive charge (+e), located in the nucleus.
    - **Electrons:** Negative charge (-e), orbiting the nucleus.
    - **Neutrons:** Neutral (no charge).
  - **Quantization of Charge:** Charge exists only in discrete packets.

$$q = n \cdot e$$

    where $e \approx 1.602 \times 10^{-19}$ C and n is an integer.
  - **Conductors vs. Insulators:** Conductors (like metals) allow charges to move freely; insulators do not.

Fundamental Electrical Phenomena III

- Charge Distribution in Conductors with Cavities Why do charges move to the outer surface?



Electrostatic equilibrium:

- The electric field inside a conductor must be zero.
- Otherwise, free charges would move.

Consequence:

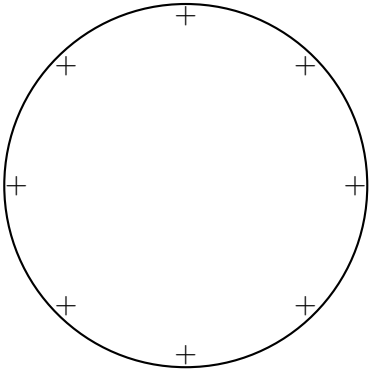
- Charges rearrange until the potential is constant.
- The conductor becomes an equipotential body.

Cavity:

- If no charge is inside, then $E = 0$ in the cavity.
- No charge can remain in the bulk.

Fundamental Electrical Phenomena IV

- Physical intuition



Intuitive picture:

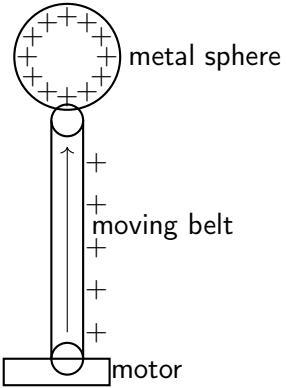
- Charges repel each other.
- They try to maximize their distance.
- The outer surface provides the largest separation.

Result:

- Charges migrate outward.
- Stable configuration: surface distribution.

Fundamental Electrical Phenomena V

- Van de Graaff Generator: Principle of Operation



Basic idea:

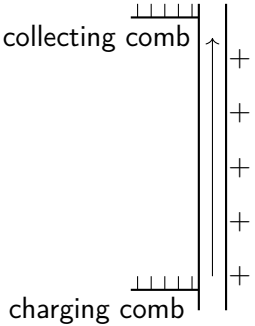
- A moving belt transports electric charge.
- Charge is continuously deposited onto a metal sphere.

Key property:

- Charge accumulates on the outer surface.
- Very high voltage can be reached.

Fundamental Electrical Phenomena VI

- Van de Graaff Generator: Charge Transport Mechanism

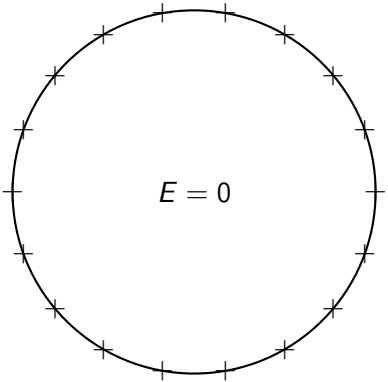


Steps:

1. A high voltage source creates charge at the lower comb.
2. The belt picks up charge by corona discharge.
3. The belt carries charge upward.
4. At the top, charge is transferred to the sphere.

Fundamental Electrical Phenomena VII

- Van de Graaff Generator: Why does voltage become very large?



Electrostatic properties:

- Inside a conductor: $E = 0$.
- Charge resides on the surface.

Consequence:

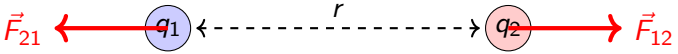
- Charge keeps accumulating on the sphere.
- Potential increases continuously.
- Limited only by air breakdown.

Fundamental Electrical Phenomena VIII

- **Coulomb's Law:** The magnitude of the electrostatic force of attraction or repulsion between two point charges is directly proportional to the product of the magnitudes of charges and inversely proportional to the square of the distance between them.

$$F = k\frac{|q_1 \cdot q_2|}{r^2},$$

where F is Electrostatic force (N), q_1, q_2 are magnitudes of the charges (C), r is distance between the centers of the charges (m), and k is Coulomb's constant $\approx 8.99 \times 10^9\,\mathrm{N} \cdot \mathrm{m}^2/\mathrm{C}^2$.



- Forces are equal in magnitude and opposite in direction (Newton's 3rd Law).
- The force acts along the line joining the two points.

Fundamental Electrical Phenomena IX

- Units and the Permittivity of Vacuum
- **Unit of Charge:** The SI unit is the **Coulomb (C)**. Defined in terms of electric current: $1\,\mathrm{C} = 1\,\mathrm{A} \cdot 1\,\mathrm{s}$.
- **Permittivity of Vacuum (ϵ_0):** Coulomb's constant k is often expressed as:

$$k = \frac{1}{4\pi\epsilon_0}$$

- **Value of ϵ_0:**

$$\epsilon_0 \approx 8.854 \times 10^{-12}\,\frac{\mathrm{C}^2}{\mathrm{N} \cdot \mathrm{m}^2}$$

- This constant represents the capability of a vacuum to permit electric field lines.

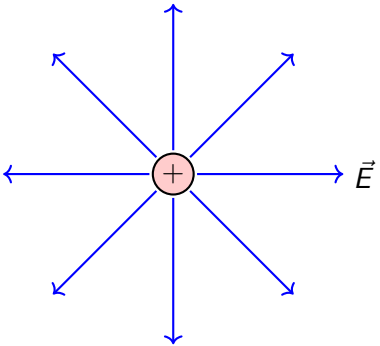
The Electrostatic Field I

- **Definition:** A region around a charged particle where a force would be exerted on other charged particles.
- **Electric Field Strength ($\vec{E}$):** Defined as the force (F) per unit positive test charge (q_0).
- **Equation:**

$$\vec{E} = \frac{\vec{F}}{q_0} \quad \text{[Unit: } N/C \text{ or } V/m\text{]}$$

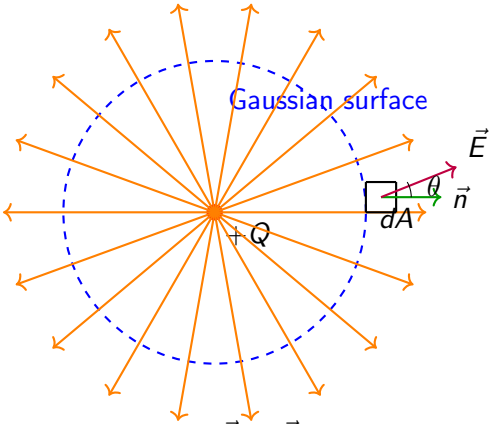
The Electrostatic Field II

- **Electric Field Lines of a Positive Point Charge**



- Electric Field Lines
  - Originate from positive charges, terminate at negative charges.
  - The density of lines represents the field strength.
  - For a **Point Charge** Q, the magnitude is: $E = k\frac{|Q|}{r^2}$
  - The field is spherically symmetric.

Gauss's Law I



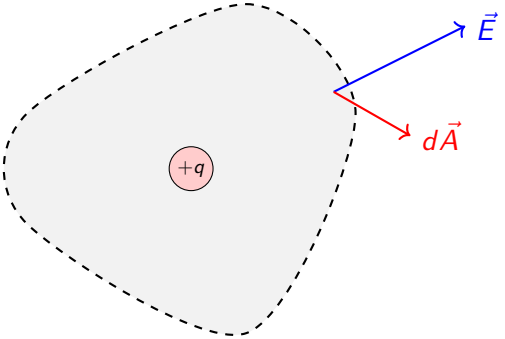
$$\Phi = \int \vec{E} \cdot d\vec{A}$$

- **Electric Flux (Φ_E):** The measure of the "flow" of the electric field through a surface.

$$\Phi_E = \oint_S \vec{E} \cdot d\vec{A}$$

Gauss's Law II

- **Flux through a Gaussian Surface**



Gaussian Surface (S)

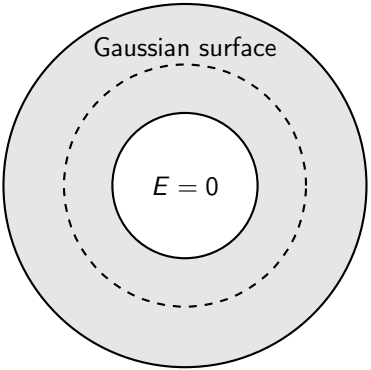
- **Gauss's Law Statement:** The net electric flux through any closed surface is equal to the net charge enclosed divided by ε_0.
- Mathematical Form

$$\oint_S \vec{E} \cdot d\vec{A} = \frac{Q_{\text{enclosed}}}{\varepsilon_0},$$

where $d\vec{A}$ is the area vector (perpendicular to the surface).
- This law is fundamental for calculating fields of symmetric objects (spheres, cylinders, planes).

Gauss's Law III

- Charge Distribution in Conductors with Cavities Why do charges move to the outer surface? Gauss's law argument



Gauss's law: $\oint \mathrm{E} \cdot dA = \frac{Q_{\text{inside}}}{\varepsilon_0}$

- Inside the conductor: $\mathrm{E} = 0$
- Therefore the flux is zero.
- Hence $Q_{\text{inside}} = 0$.

Conclusion:

- No net charge can exist inside the conductor.
- All excess charge must reside on the surface.

The End

Thank you for your attention!

Acknowledgements Thank you for the course materials for Szilárd ZSÓKA and Zoltán VARGA.